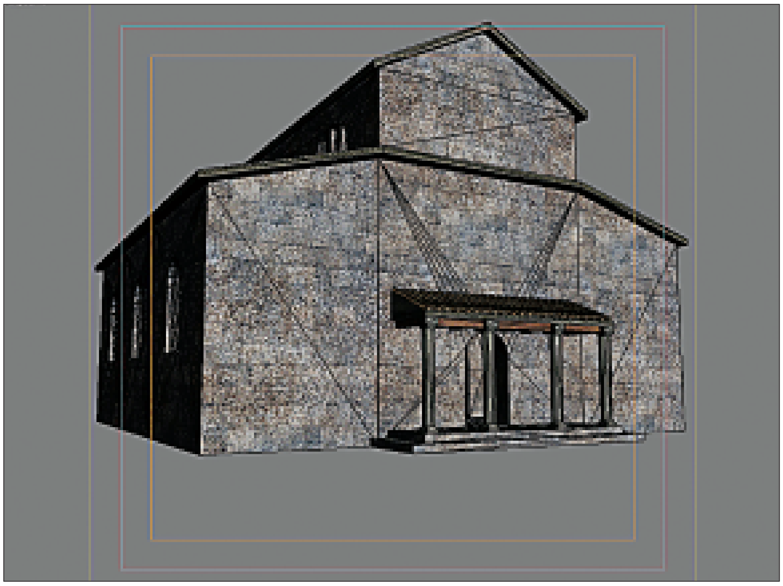
very labour-intensive process. Naturally, short-cuts were used as often as possible, and several *cels* would be combined together to create a scene. A cel is a piece of transparent material, usually about a foot across, on which elements of the scene are drawn or painted. In the simplest example, say, for a ball rolling across a floor, a background scene would be painted, and then several cels representing the ball in different positions would be placed one after the other on top of the background, and then photographed. Although we now mainly use computers for animation, terminology that comes from those early days, such as the word cel, is often found in use with modern systems. There is also a healthy market for used cels — your favourite Disney or Anime character makes for an interesting collector's item.

3D started to enter the animation industry in the 1950s with the production of Gumby. This featured a rather odd green character together with his red pony, developed by the American, Art Clokey. This production used clay figures, the positions and gestures of which were changed between each shot. Again, this is very labour intensive, with the small changes between individual frames being made by hand and then again individually photographed. Although one might consider this an out-dated system, of little value today, in recent years this technique has been perfected with enormous success by Nick Park in his *Wallace and Gromit* films, such as *The Wrong Trousers* and *The Curse of the Were-Rabbit*.

And then along came computers. Although the cathode ray tube (CRT) is dated back to 1885, it was not until the 1950s that these were connected to early computers in order to create images on the screen. The term "computer graphics" was coined in 1960 by Boeing's William Fetter, with the first primitive video game, *Spacewars*, developed the following year at MIT. The next couple of decades saw enormous development, and it is interesting to look at modern systems and realise that many of the advanced techniques we are using today, in modern CPU architectures and graphics systems, were in fact developed back in the 60s and 70s. In those days many of these developments were largely theoretical, and when practical, were used on very expensive and large machines. Modern computer technology has brought these innovations truly to life.

Graphics on a computer fall into two distinct types of image: raster- and vector-based images. If you scan a page of text or a drawing, or if you take a digital photograph, you create a raster image. The word raster refers to the grid-like structure that makes up the image. This is a number of rows consisting of individual picture elements (pixels). Each pixel is basically a dot (perhaps a small square might be a better description) effectively of uniform colour. The number of these pixels in each row and the number of rows gives the resolution of the image — a representation of the amount of detail held. The important point is that the amount of detail is limited — if you zoom in on one part of the image, eventually you will see only individual pixels.

Vector images are very different. With these, the image is held in a mathematical format. A coordinate system is used to represent position within the image, and this is used to define points and lines — a line simply connects two points. Other more complex geometric structures are simply combinations of lines — a circle for instance, is a large number of short lines connected together; the greater the number, the smoother the circle.



The top image shows a basilica from the online game *Roma Victor* in the 3D Studio editing window. Below, the same after rendering